

Practice Exercise – 2026-05-20

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(a)

2count – Invalid – variable name cannot start with a digit

item count – Invalid – variable name cannot include spaces

_itemCount – Valid

(b)

2count → count

item count → item_count

```
int : 4 byte(s)
double : 8 byte(s)
char : 1 byte(s)
```

(c)

It prints the maximum number the data types int and double can hold.

```
int max: 2147483647
dbl max: 1.797693e+308
int min: -2147483648
flt max: 3.402823e+38
```

(d)

They get removed.

```

# 6 "main.c"
int main() {
    int count = 10;
    int item_count = 20;
    int _itemCount = 30;

    int age = 23;
    double gpa = 3.856712345;
    char grade = 'A';

    printf("int : %zu byte(s)\n", sizeof(int));
    printf("double : %zu byte(s)\n", sizeof(double));
    printf("char : %zu byte(s)\n", sizeof(char));

    printf("int max: %d\n", 0x7fffffff
# 21 "main.c"
    );
    printf("dbl max: %e\n", ((double)1.79769313486231570814527423731704357e+308L)
# 22 "main.c"
    );
    printf("int min: %d\n",
# 23 "main.c" 3 4
        (-0x7fffffff - 1)
# 23 "main.c"
    );
    printf("flt max: %e\n", 3.40282346638528859811704183484516925e+38F
# 24 "main.c"
    );

    printf("%10d\n", age);
    printf("%+10d\n", age);
    printf("%010d\n", age);

```

(e)

```

adrp    x0, .LC0
add     x0, x0, :lo12:.LC0
bl      printf

```

(f)

10d → prints the value of age with a width of 10 characters. It is right aligned.

−+10d → prints the value of age with a width of 10 characters and its sign. It is left aligned.

010d → prints the value of age with a width of 10 characters. The whitespace at the start is filled with 0s.

.3f → prints the value of gpa up to 3 decimal places

E → prints the value of gpa in scientific notation

c = d → first prints the character assigned to grade and then prints its ASCII value

```
          23
+23
0000000023
3.857
3.856712E+00
A = 65
```